

Traumatic Timebomb?: A case of bifurcating coronary artery dissection after motor vehicle collision

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Introduction

- Syncope is a common presentation requiring a broad differential and multidisciplinary evaluation.
- It may represent the sequela of underlying pathology, while simultaneously acting as a precipitating event for other complications (i.e., traumatic injuries).

Case Presentation

- 56-year-old male with history of recurrent syncope (not previously evaluated); no other known PMH or complaints
- Admitted after witnessed syncopal event as restrained driver resulted in MVC; cleared by trauma survey
- Initial work up:
 - mildly elevated, down-trending hs-troponin (peak 69 ng/L)
 - nonischemic serial ECGs; no events on >48 hrs of tele
 - Unremarkable TTE
 - Coronary CTA: severe proximal LAD stenosis (CAC 18)
- Procedural complication:
 - Non-emergent coronary angiography confirmed a 90% proximal LAD lesion, prompting PCI
 - Immediately after pre-dilation with SC balloon, a large spiraling dissection was noted, extending from the mid LAD into the LM and LCx; confirmed with IVUS
 - Emergent culotte LM-LAD-LCx bifurcation PCI with three overlapping DES was performed successfully
- Postprocedural TTE revealed new systolic dysfunction (EF 30-35% with WMA c/w infarct in LAD territory), for which GDMT was initiated.
- While observed in CICU for post-procedural monitoring, he suffered a 24-second asystolic pause preceded by progressive P-P prolongation, prompting dual chamber PPM implantation.

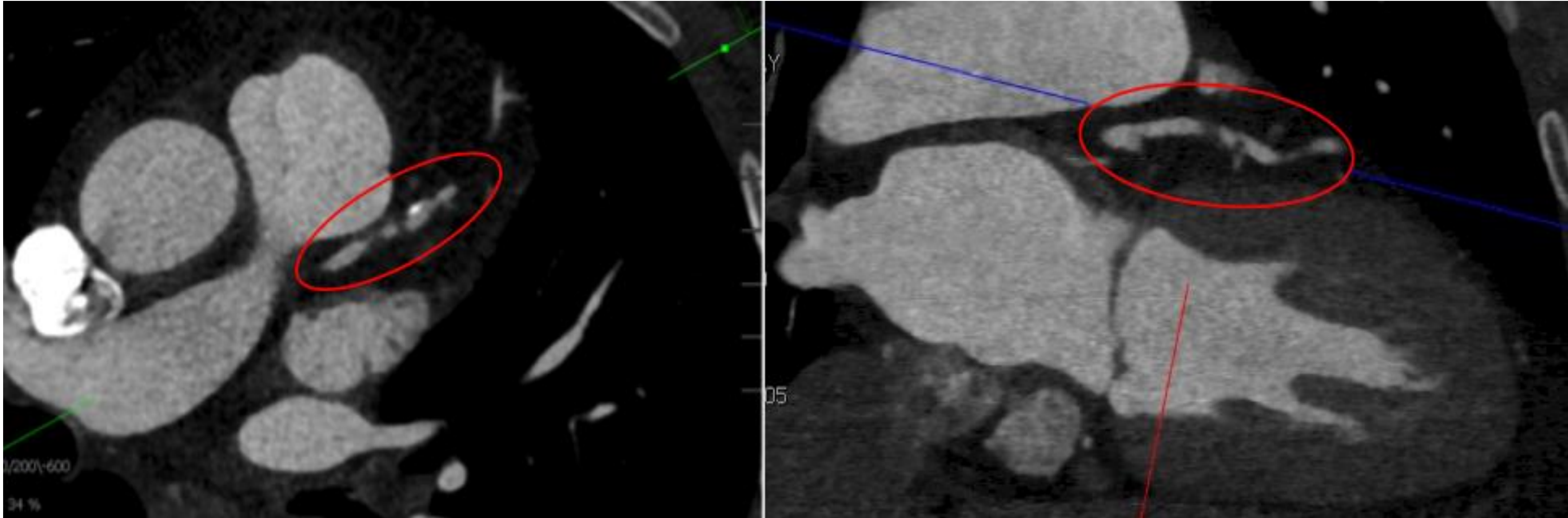


Figure 1: Orthogonal images from coronary CT angiography demonstrating a severe, noncalcified lesion in the proximal LAD (red circle)

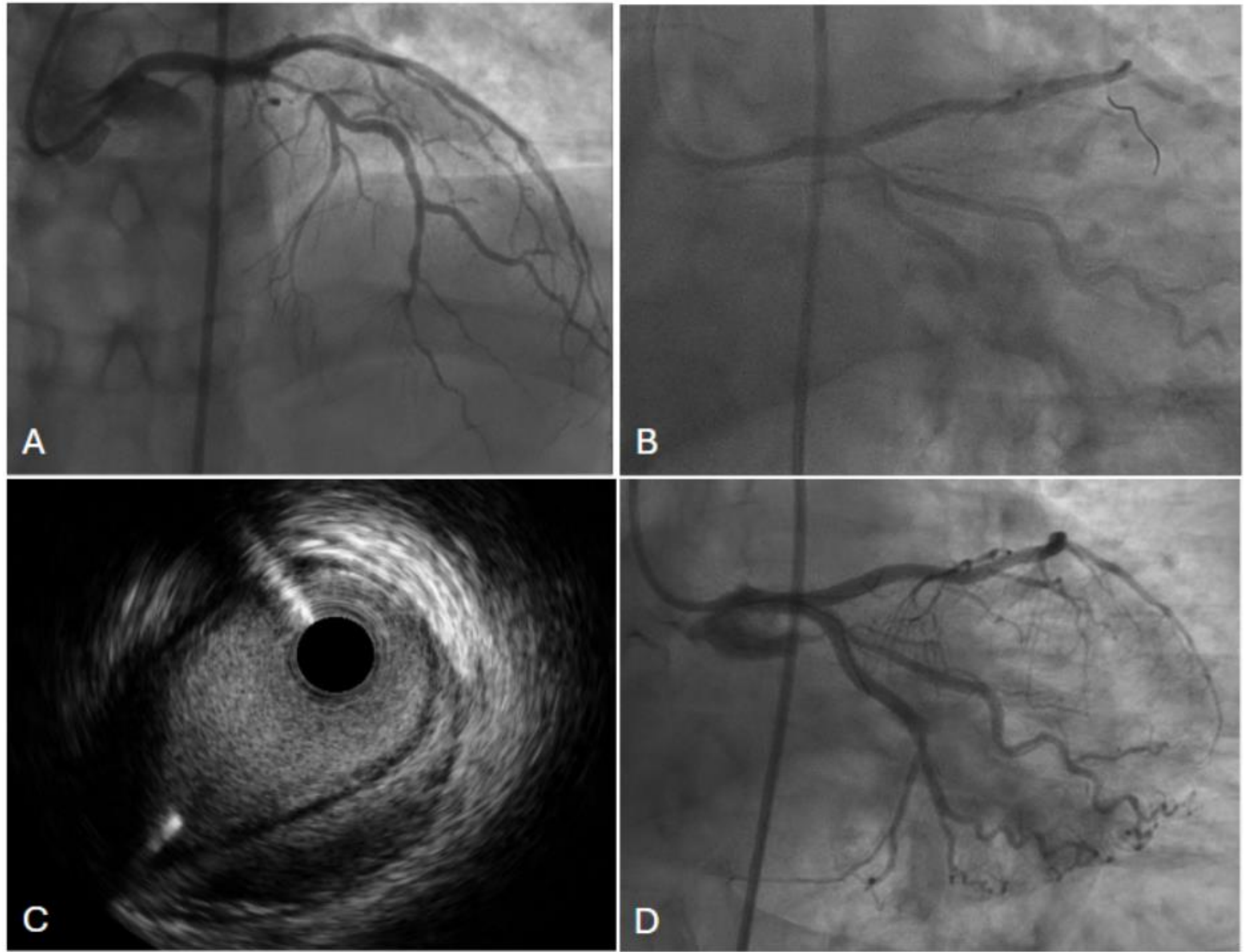


Figure 2: Coronary angiograms displaying (A) the initial shot of the left system prior to PCI, (B) the spiraling dissection from mid LAD back toward LM and down LCx, and (D) the final angiographic result post culotte LM-LAD-LCx bifurcation. (C) IVUS still image demonstrating dissection flap and false lumen at bifurcation.

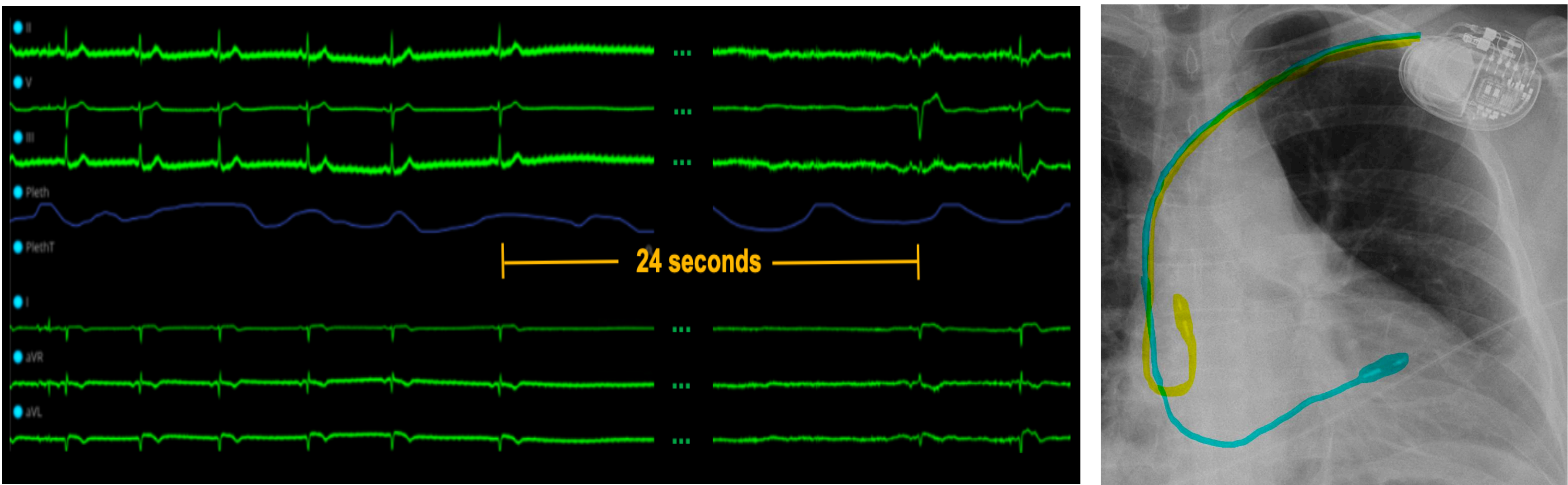


Figure 3: (A) Telemetry strip capturing 24 sec sinus arrest, prompting insertion of dc-PPM. (B) CXR with LBB (cyan) and Bachmann's bundle (yellow) PPM leads

Discussion

- Our patient's history of recurrent syncope was most likely secondary to severe sinus node dysfunction.
 - However, a transient ischemic ventricular arrhythmia preceding his MVC cannot be entirely ruled out.
- The high-grade, noncalcified, single vessel coronary lesion revealed on CCTA most likely represented a traumatic vascular injury following air bag deployment.
 - Supported by lack of preceding exertional symptoms and initial admission work-up not consistent with ACS.
- The periprocedural coronary artery dissection was most likely a delayed propagation, triggered by balloon expansion, from an initial traumatic insult to the vascular wall suffered during his MVC.
 - If not intervened on, this patient's severe LAD lesion may have otherwise propagated spontaneously (potentially leading to SCD, in the extreme).

Key Points

- This case demonstrates how a common clinical presentation can involve challenging management of uncommon underlying pathophysiology.
- Traumatic coronary artery dissections can result from deceleration & shearing forces during a MVC, even when other severe traumatic sequelae are absent.
- This case also highlights the critical importance of multidisciplinary care team evaluation to avoid MACE.

References

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